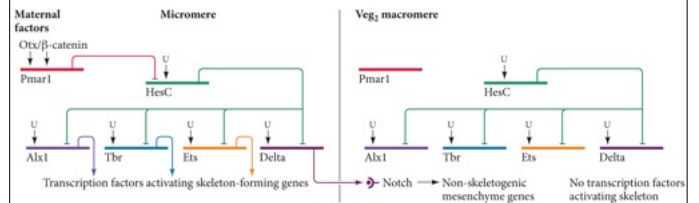


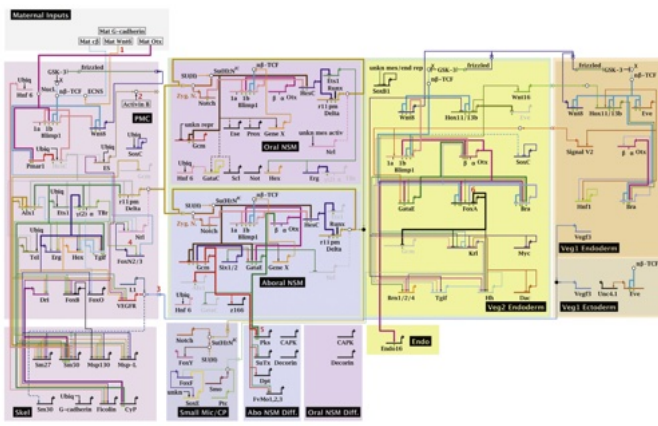
Examples of how GRNs are modified during evolution

True or False?

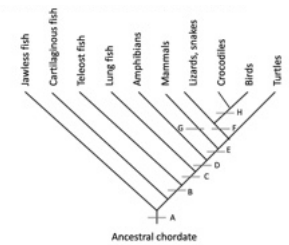


- ☐ Micromeres express high levels of HesC.
- ☐ Knock down of Pmar1 or HesC in micromeres will result on the same phenotype.
- ☐ Injection of dishevelled (DSH) in *Veg2* macromeres will result in the formation of ectopic skeleton.
- ☐ *Veg2* macromeres cultured in isolation are able to form non-skeletogenic mesenchyme (SMCs).
- ☐ Knocking down the HesC gene in isolated *Veg2* macromeres will cause them to give rise to skeletal cells.

“Development of the body plan is controlled by large networks of regulatory genes”

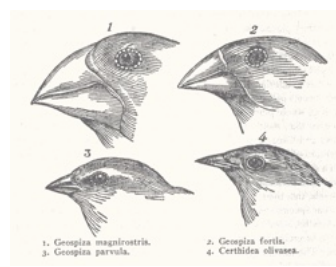


How do GRNs inform our understanding of evolution?

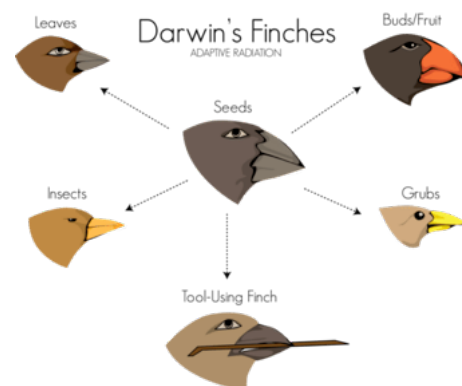


- Organisms become complex **not** because of increased number or complexity in genes but because of increased complexity of genetic regulatory networks controlling development
- Evolution not only modifies genes, but modifies “genetic switches”, allowing for huge variety of possibilities of gene-expression patterns.

Example 1: Darwin’s Finches:



Beak morphology vs. feeding behavior



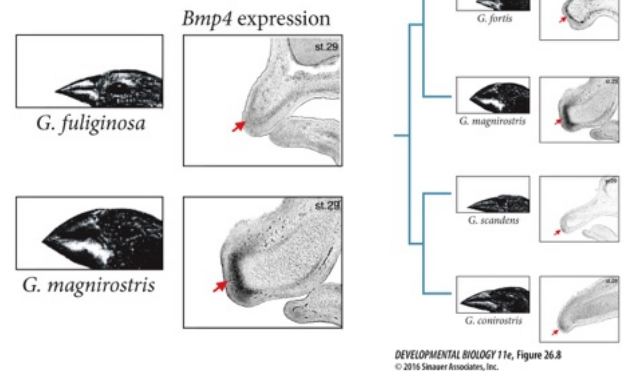
- What caused diversity of finches' beaks?

- **Traditional Darwinian answer:**

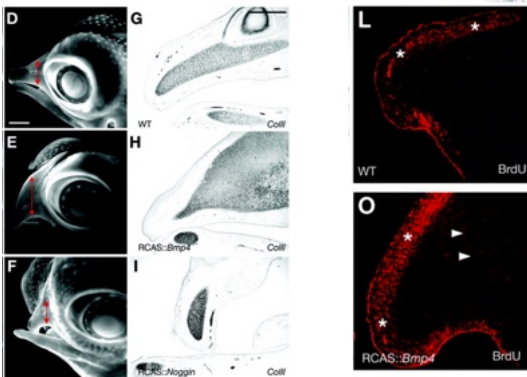
Gradual process of adaptation via mutations, recombinations of genes

Result: Different finches have significantly different genomes

Finch beak morphology is correlated with BMP4 expression domains



Manipulation of BMP activity alters amount of beak mesenchyme



- **Possible Evo-Devo answer:**

No major changes in "structural" genes

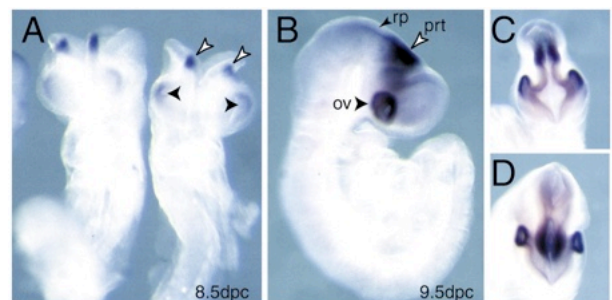
Instead, changes in regulation of **BMP4** gene, itself a regulatory gene that controls beak size and shape by regulating other genes that produce bones

- The more strongly **BMP4** is expressed during development, the larger and stronger the beak.

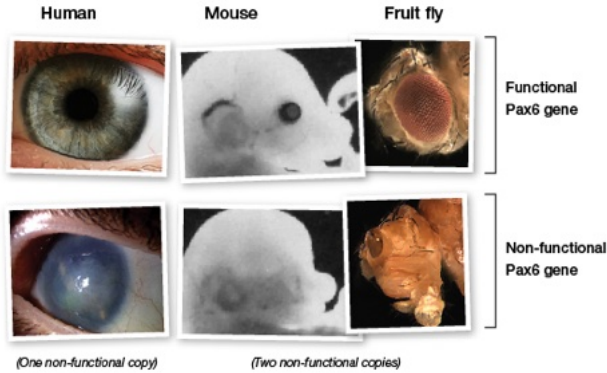
Example 2: what happened to the eyes of the blind cavefish?



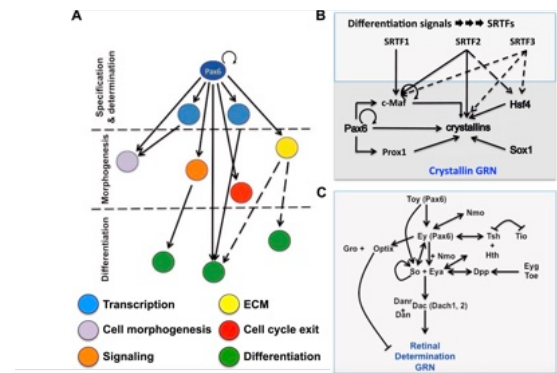
Pax6 is a gene that is expressed at the developing eye during embryonic development



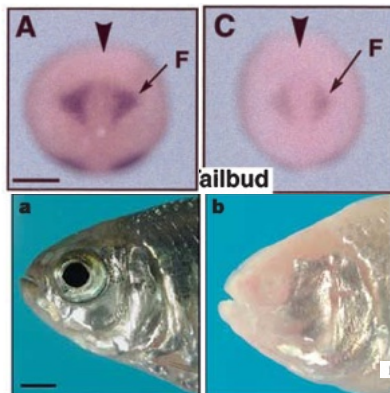
Mutation of Pax6 causes eye defects in vertebrates and invertebrates



PAX6 is a crucial node in the GRN that controls eye formation

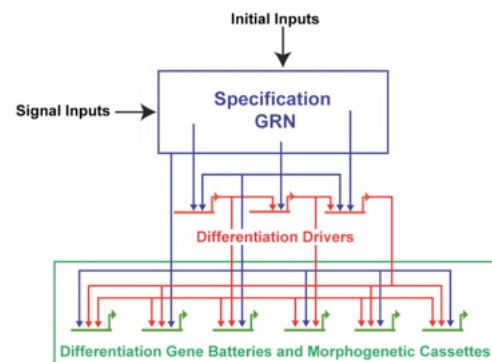


Cavefish have low levels of Pax6 expression in the optic cups!



Journal of Heredity 2005;96(3):185-196
Dev Genes Evol (2001) 211:138-144
 NATURE | VOL 431 | 14 OCTOBER 2004

Thus, it is not the accumulation of mutations that drive evolution, but rather how they modify developmental pathways



PART III:

Genome Architecture and Function

Eukaryotes

- **Complex gene structure**
- **Large genomes (0.1 to 10 billion bp)**
- **Exons and Introns (interrupted)**
- **Low coding density (<30%)**
 - 3% in humans, 25% in Fugu, 60% in yeast
- **Alternate splicing (40-60% of all genes)**
- **High abundance of repeat sequence (50% in humans) and pseudo genes**
- **Nested genes: overlapping on same or opposite strand or inside an intron**

genome = DNA content of a complete haploid set of chromosomes
 = DNA content of a gamete (sperm or egg)

Species	Chromosomes	cM	DNA content/ haploid (Mb)	year sequence completed	genes/ haploid	genes have introns?
<i>E. coli</i>	1	N/A	5	1997	4,200	no
<i>S. cerevisiae</i>	16	4000	12	1997	5,800	rarely
<i>C. elegans</i>	6	300	100	1998	19,000	nearly all
<i>D. melanogaster</i>	4	280	180	2000	14,000	nearly all
<i>M. musculus</i>	20	1700	3000	2002 draft 2005 finished?	22,500?	nearly all
<i>H. sapiens</i>	23	3300	3000	2001 draft 2003 finished	22,500?	nearly all

Note: cM = centi Morgan = 1% recombination
 Mb = megabase = 1 million base-pairs of DNA
 Kb = kilobase = 1 thousand base-pairs of DNA

SPECIAL SECTION

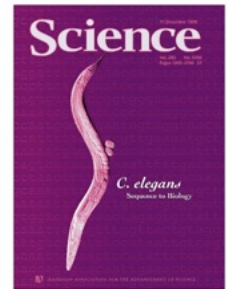
C. ELEGANS: SEQUENCE TO BIOLOGY

Genome Sequence of the Nematode *C. elegans*: A Platform for Investigating Biology

The *C. elegans* Sequencing Consortium*

- The first multicellular organism genome sequenced
- 97 Mb
- over 19,000 genes (~ 5000 essential)
- The 97-Mb sequence is a composite of

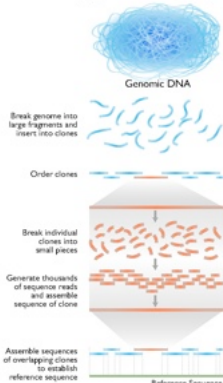
- 2527 cosmids,
- 257 YACs,
- 113 fosmids, and
- 44 PCR products



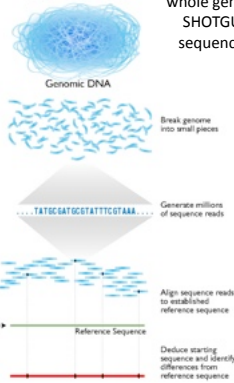
Science 282, 2012 (1998)

Human Genome Sequencing

Generating a Reference
Genome Sequence
(e.g., Human Genome Project)

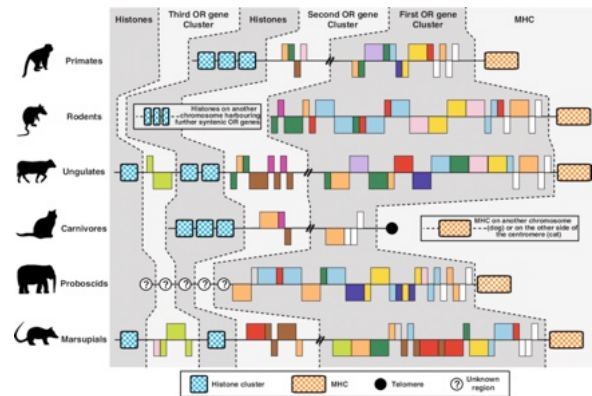


Generating a Person's
Genome Sequence
(e.g., Circa ~2016)

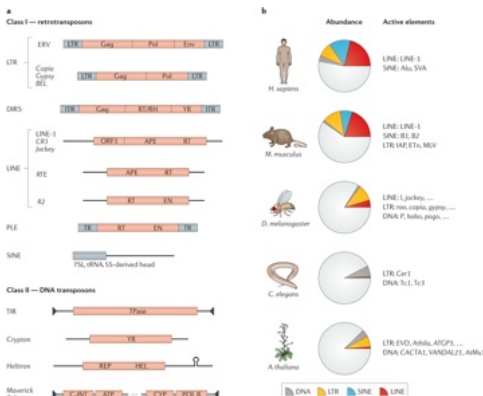


This is called
whole genome
SHOTGUN
sequencing

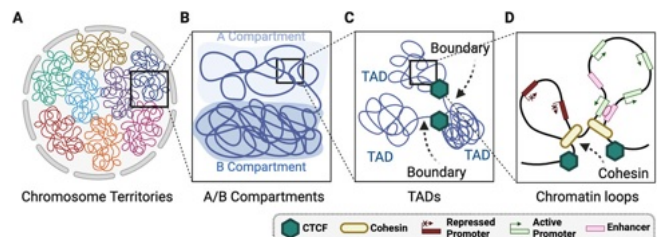
This has allowed for whole-genome comparison
between species AND individuals



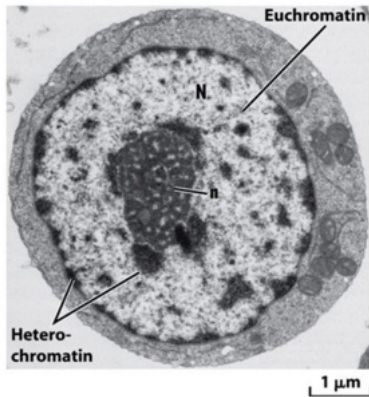
Abundance of transposable elements in metazoan genomes



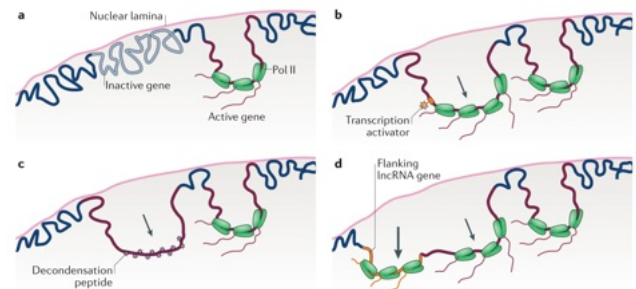
But what about the 3D organization of the
genome?



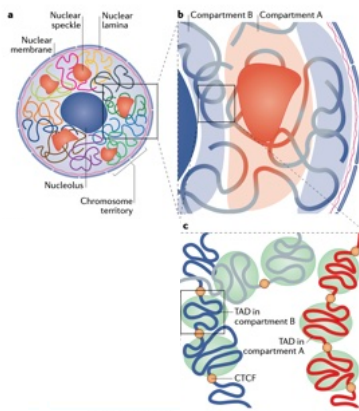
Chromatin organization: heterochromatin vs. euchromatin



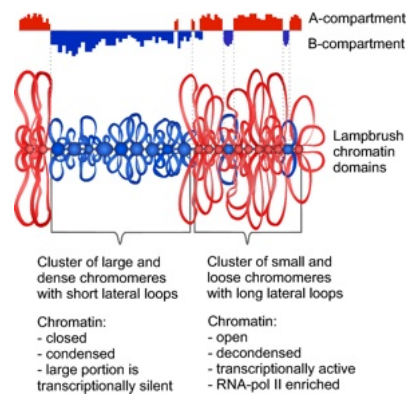
But what about the organization of the genome? *Studying the epigenome*



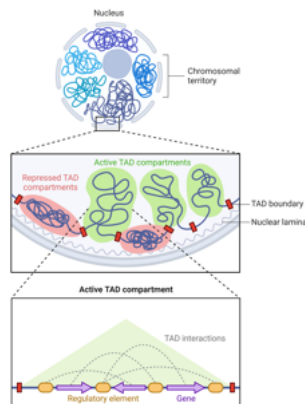
Organization of the genome



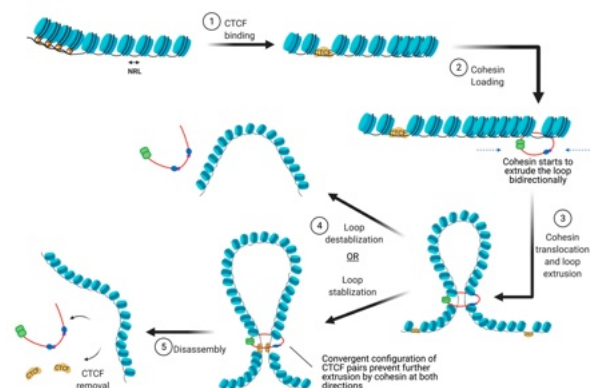
A and B compartments:



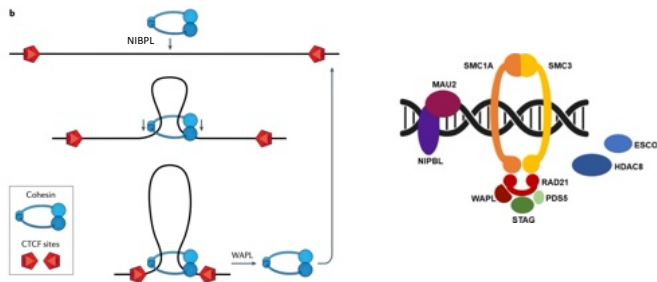
What are TADs?



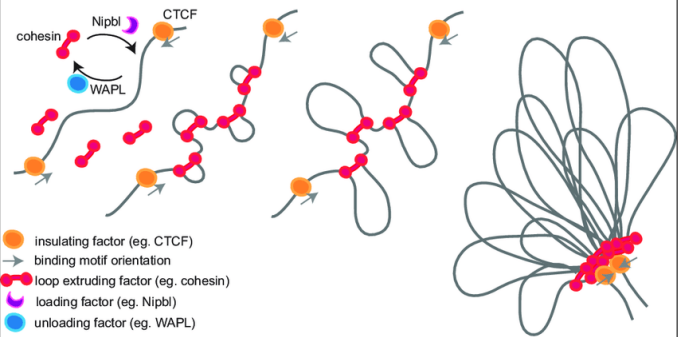
The loop extrusion model



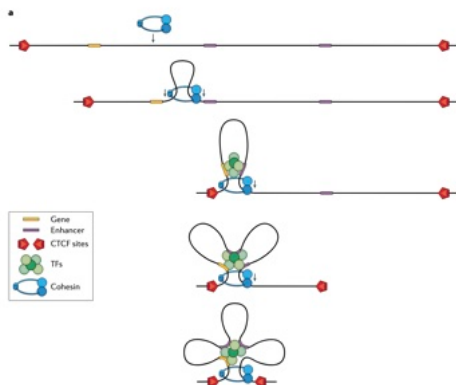
CTCF and Cohesin (and WAPL, NIBPL)



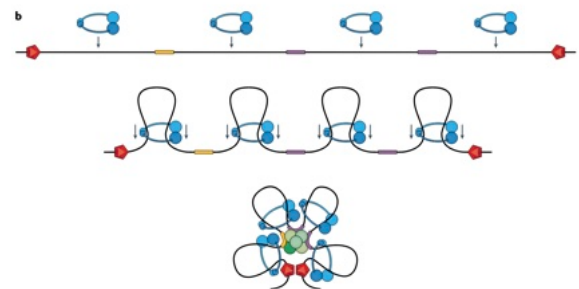
Loop extrusion model in TAD formation



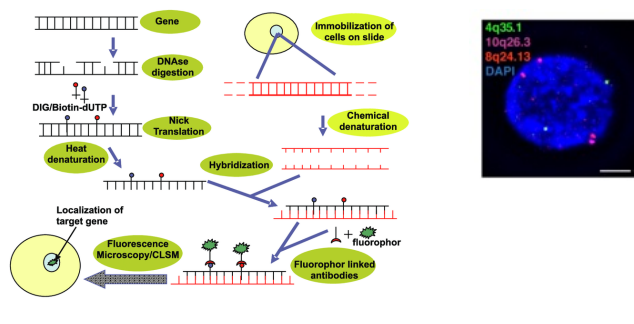
Coupling gene expression and loop formation



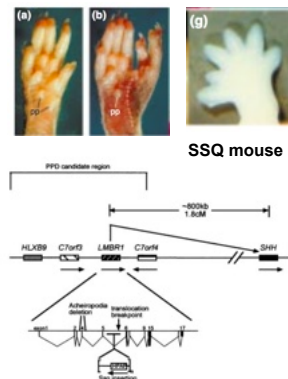
Coupling gene expression and loop formation



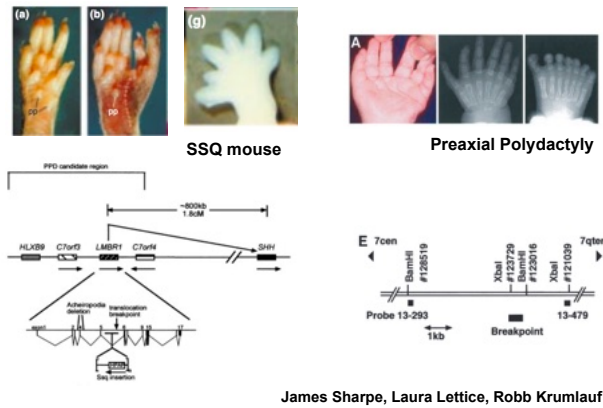
How to detect DNA-DNA interactions?



Identification of ZRS enhancer:



Identification of ZRS enhancer:



Identification of ZRS enhancer:



Ernest Hemingway's Cats

But there are additional signaling systems in the limb:
ZPA: Zone of Polarizing Activity



Figure 19.2

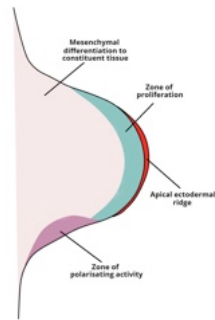


Figure 19.20

Sonic hedgehog (*Shh*) is localized to the ZPA (arrows)

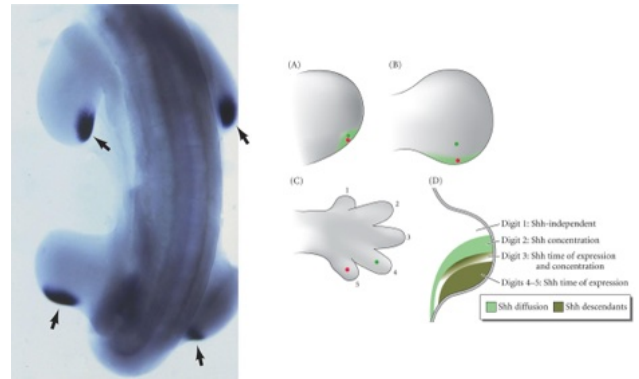
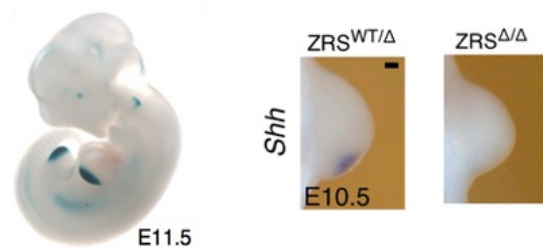
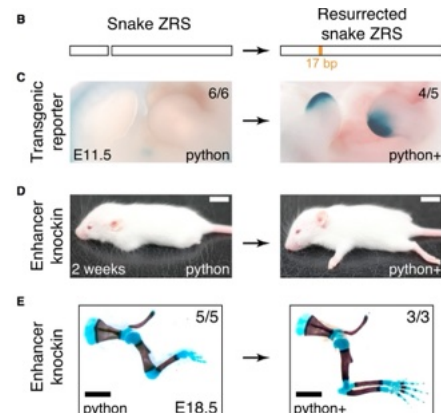


Figure 19.20

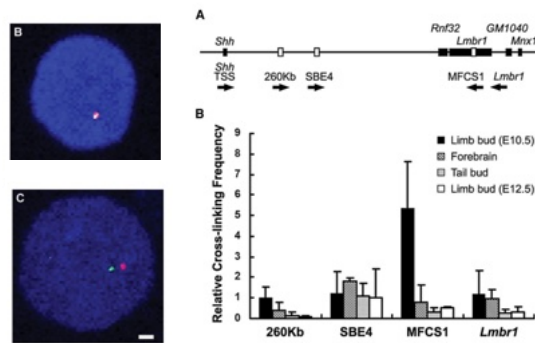
Identification of ZRS enhancer:



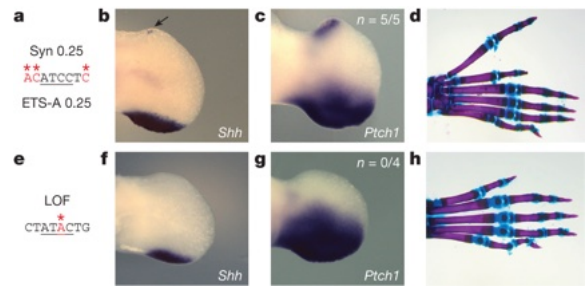
Identification of ZRS enhancer:



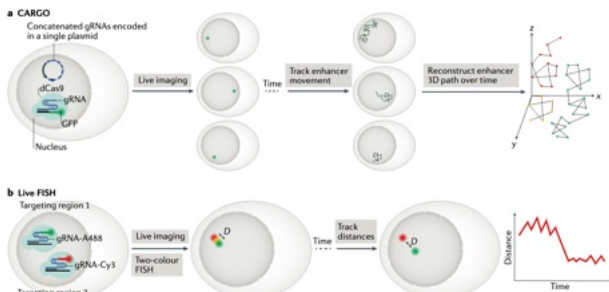
Identification of ZRS enhancer with DNA FISH



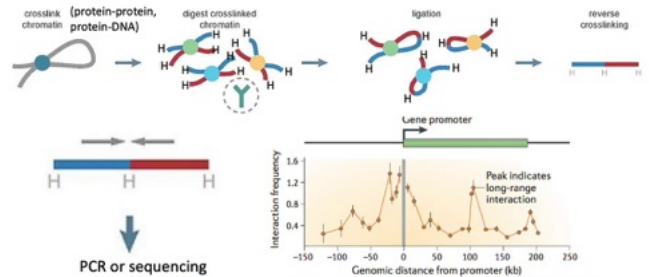
Identification of ZRS enhancer with DNA FISH



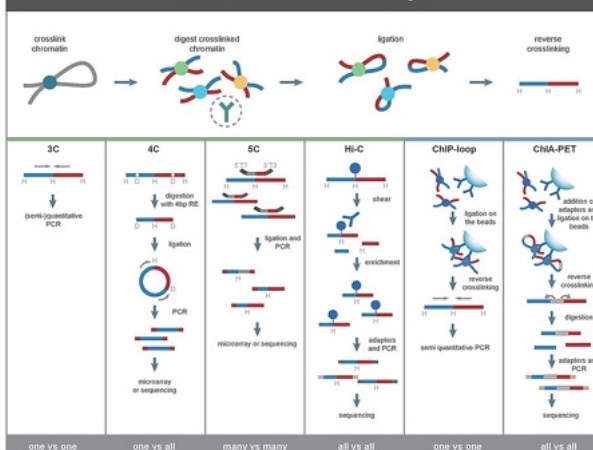
Updating DNA FISH:



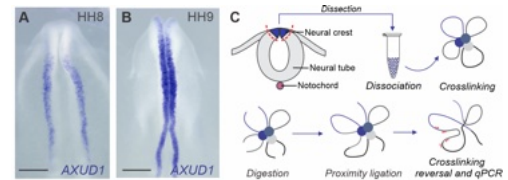
Chromatin conformation capture experiments



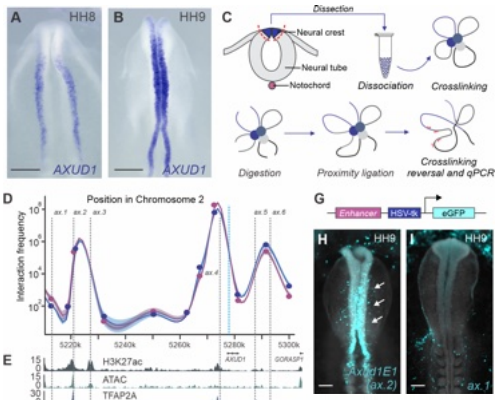
Chromosome Conformation Technologies



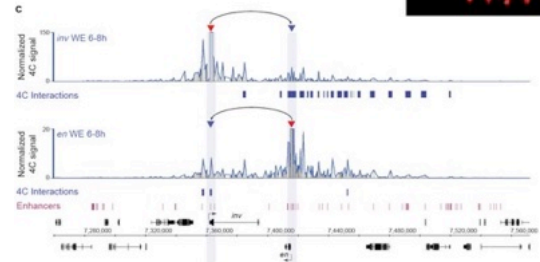
Chromatin conformation capture experiments



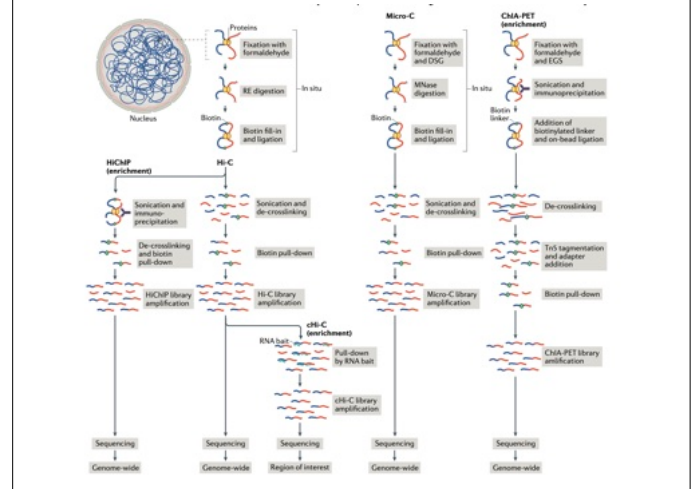
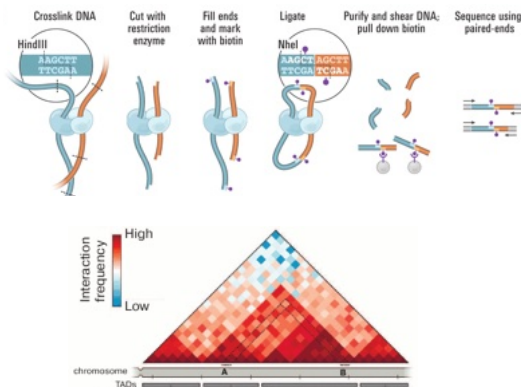
Chromatin conformation capture experiments



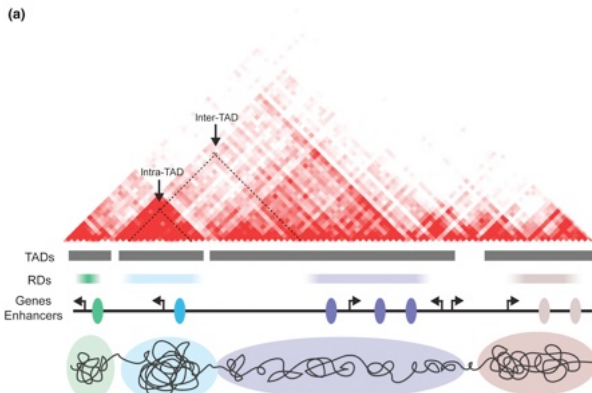
Results from a 4C experiment



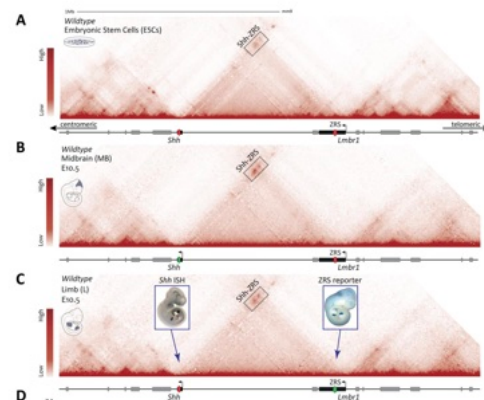
Assembling the genome connectomes with Hi-C



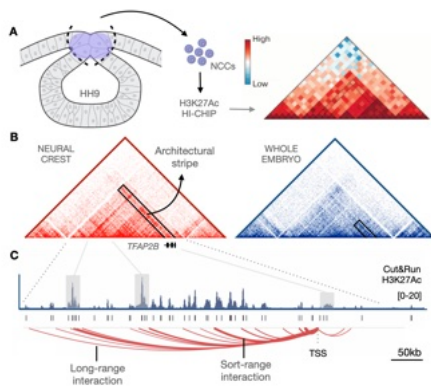
Results from a Hi-C experiment



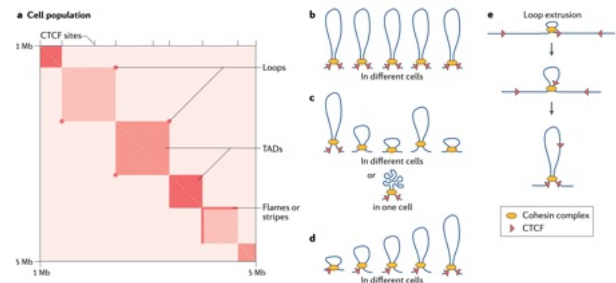
Results from a Hi-C experiment



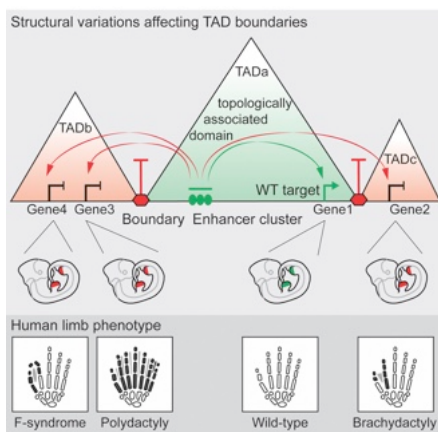
Results from a Hi-C experiment



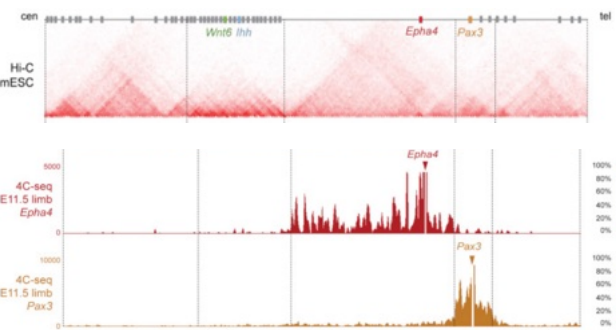
Interpreting Hi-C/Hi-ChIP results:



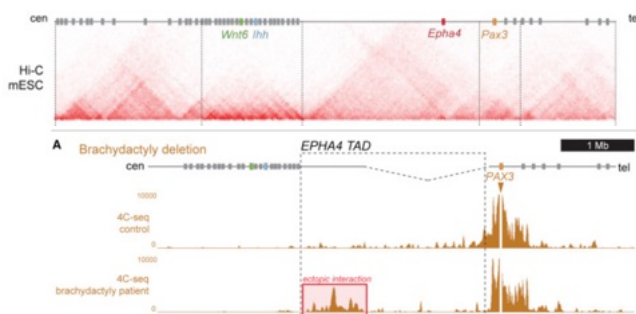
Why is this important?



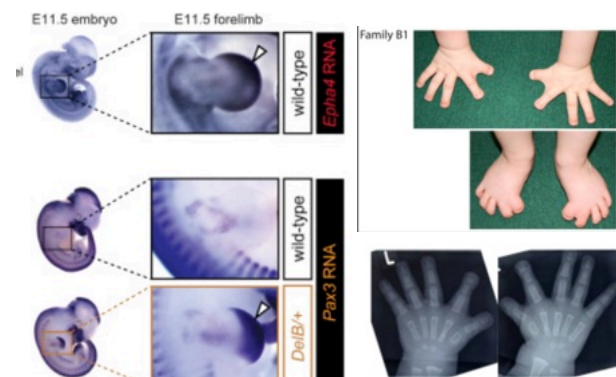
Results from a Hi-C experiment



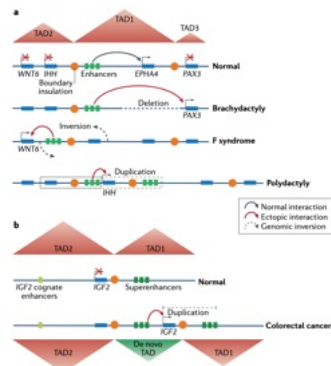
Results from a Hi-C experiment



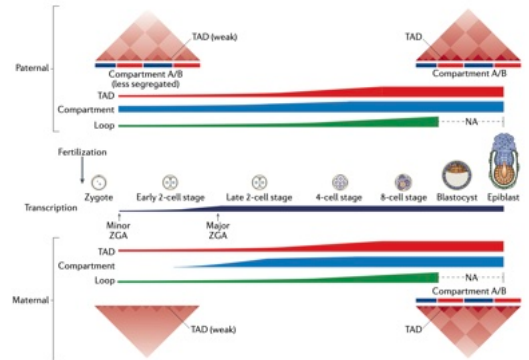
Results from a Hi-C experiment



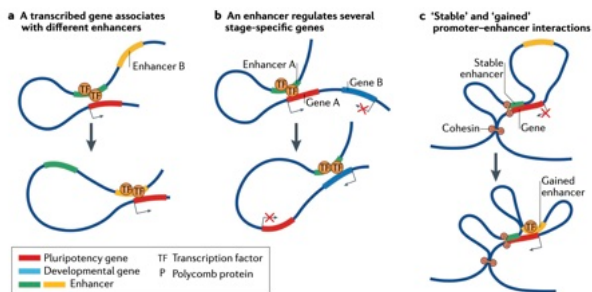
Changes in TAD structure results in misregulation of genes



When are compartments formed during development:



Types of intra-tad DNA loops



Single cell Hi-C: is chromatin organization heterogeneous within a cell population?

